

A stochastic explanation for observed local-to-global foraging states in *Caenorhabditis elegans*

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eLife Assessment

This **useful** paper uses a quantitative modeling approach to explore a putative mechanism underlying a well-studied behavioral transition in the nematode *C. elegans*. The premise, that what has been considered a two-state behavior can instead be described as a process whose parameters are smoothly modulated within a single state, is intriguing. However, in the paper's current state, concerns about the model and its fit to empirical data make the support for this idea **inadequate**.

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Abstract

Abrupt changes in behavior can often be associated with changes in underlying behavioral states. When placed off food, the foraging behavior of *C. elegans* can be described as a change between an initial local-search behavior characterized by a high rate of reorientations, followed by a global-search behavior characterized by sparse reorientations. This is commonly observed in individual worms, but when numerous worms are characterized, only about half appear to exhibit this behavior. We propose an alternative model that predicts both abrupt and continuous changes to reorientation that does not rely on behavioral states. This model is inspired by molecular dynamics modeling that defines the foraging reorientation rate as a decaying parameter. By stochastically sampling from the probability distribution defined by this rate, both abrupt and gradual changes to reorientation rates can occur, matching experimentally observed results. Crucially, this model does not depend on behavioral states or information accumulation. Even though abrupt behavioral changes do occur, they are not necessarily indicative of abrupt changes in behavioral states, especially when abrupt changes are not universally observed in the population.

Introduction

The search for food in the absence of informative sensory cues is an essential animal behavior¹. A foraging strategy that is observed in nearly all animals is the Area Restricted Search (ARS)². In ARS animals randomly forage for food³, but appetitive sensory cues (like an encounter with food) will cause the animal to restrict their search area by reorienting more frequently, thus increasing the likelihood of food encounters. Conversely, when removed from food, animals will decrease their reorientation rate to increase dispersal^{1,4}. *Caenorhabditis elegans* and *Drosophila melanogaster* larva appear to progressively increase their diffusion constant while foraging off of food by decreasing their rates of reorientation⁴. However in separate studies^{5,6}, individual worms appear to make an abrupt change from a high to low rate of reorientation. This behavior has been described as a switch from a local to global search strategy, that relies on evidence accumulation to trigger the behavioral switch⁶. In López-Cruz *et al.*⁵ the foraging behaviors of individual worms were tracked for 45 minutes after being removed from food. As observed previously⁴, the reorientation rate from this study followed an exponential decay (**figure 1a**). It was reported that roughly half the worms appeared to make a sudden single switch from local to global search as observed in Calhoun *et al.*⁶, however the other half appeared to produce no discernable change in search strategy, or exhibited multiple switches (**figure 1 b-d**). Whether or not a worm performed a single decision was defined in Lopez-Cruz *et al.*⁵ by fitting individual reorientation data to two lines using the MATLAB function *findchangepts* (**figure 1b**).⁷ This function divides each trace into two regions that are defined by minimizing the sum of the residual squared error of two local linear regressions. The location of the transition point is varied until the total residual error attains a minimum (**figure 1b**). A large change in slope indicates a sudden change in reorientation rate, and the intersect between the two lines determines the decision-time⁵ (**figure 1b**). However, the resulting distributions of slope-differences ($s1-s2$) and transition times for all of the experimental data are continuous, with no clear boundary between deciders and non-deciders (**figure 1e**).

Why do some worms appear to make a decision, while others do not? In aggregate, the reorientation rate decays (**figure 1a**). Despite the population average conforming to a gradual decay, individual trajectories produce a wide diversity of trajectories which sometimes conform to an apparent drop in reorientation rate (**figure 1b**), while others do not (**figure 1c**). If the worms are executing a decision, this would seem to indicate only a fraction of the worms decide to switch from local to global foraging strategies, while others perform an alternative strategy.

A central challenge with defining these possible state transitions is the stochastic nature of the behavior itself. Reorientations are random and follow Poisson statistics⁸⁻¹⁴. This is very similar to the temporal behavior of individual molecules in solution¹⁵. Individual molecules defined by the same reaction kinetics can stochastically produce long or short time intervals between reaction events, not because one molecule is inherently faster than the other, but because they are stochastically sampling from the same probability distribution. The diversity of times between individual worm reorientations is very similar to this. The exponentially decaying reorientation rate emerges from the average of trajectories that do not necessarily conform to this curve individually. However, even those that produce abrupt switches in reorientation rates could still emerge from a simple exponential decay strategy. Since reorientations occur stochastically, the abrupt changes in reorientation rates could simply be the result of stochastic sampling of an underlying decay phenomenon¹⁶.

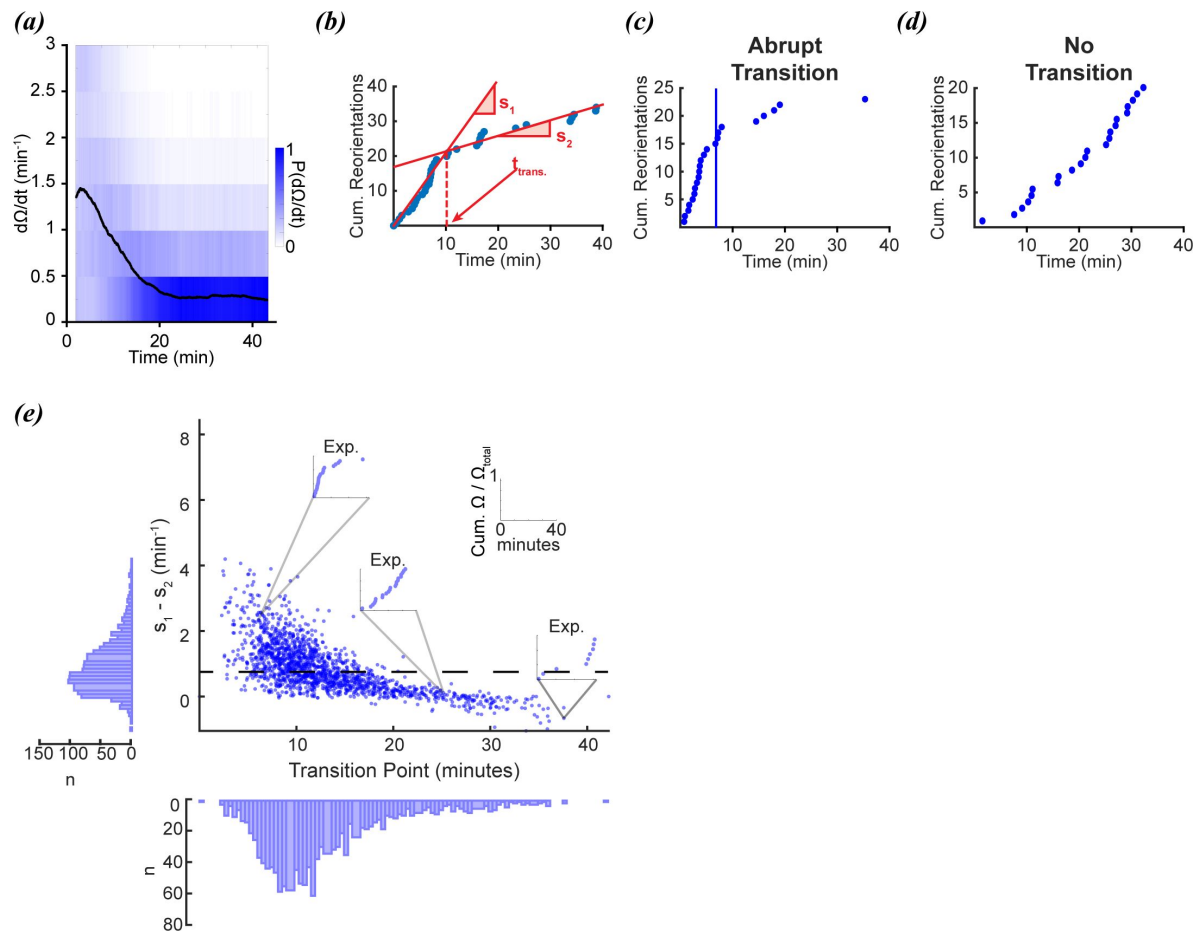


Figure 1

Foraging kinetics of *C. elegans*

(a) Average experimental population reorientation rate (black line) in a rolling 2-minute window. Blue bins represent probability of observed reorientation rate. (b) Abrupt transitions were identified by performing two linear regressions on observed reorientation curves. Transition times were defined by the intersection of the regressions. (c) An example of an experimental reorientation curve with an abrupt reorientation transition. (d) An example of an experimental reorientation curve that lacked an abrupt reorientation transition. (e) Distribution of slope differences and transition times from regressions fit to the experimental data. Insets are individual examples of experimental cumulative reorientation curves. Number of worms (N) = 1631. Dashed line represents the median slope difference. All data curated from López-Cruz *et al.* <https://doi.org/10.1016/j.cub.2019.08.001>.

Results

We tested this hypothesis by modeling individual worms by stochastic sampling of a decaying reorientation rate with the Gillespie algorithm (**figure 2a** [↗](#)), a common strategy used to model the kinetics of individual molecules¹⁷ [↗](#). With this strategy, the time between chemical events is modeled by randomly sampling from the time-interval distributions defined by the reaction rates. Although the algorithm was originally developed to model discrete molecular events based on known kinetic parameters, it can be used to generate time trajectories for any discrete events when the kinetics are known. A behavioral example of this is the Lotka-Volterra predator-prey competition model where predator and prey populations fluctuate out of phase due to predation. Stochastic fluctuations of predator and prey populations can be modeled using the Gillespie algorithm¹⁸ [↗](#)–²⁰ [↗](#).

In our model, there are two rates drawn from fitting an exponential decay function to the experimental average reorientation rate (**figure 2b** [↗](#)):

$$\frac{d\Omega}{dt} = \alpha e^{-\gamma t} \quad (1)$$

Where Ω is reorientations α is the initial reorientation rate at $t=0$, and γ is the reorientation decay rate. To model both the reorientations, and the decay in the reorientation rate, we modeled two kinetic processes, a reorientation rate (a_1) and a decaying rate of parameter M (a_2) that the reorientation rate relies on:

$$a_1 = \frac{d\Omega}{dt} = \frac{\alpha}{M_0} M(t) \quad (2)$$

$$a_2 = \frac{dM}{dt} = -\gamma M \quad (3)$$

therefore:

$$a_1 = \alpha e^{-\gamma t} \quad (4)$$

where M_0 is the amount of M at $t = 0$. In this model, the reorientation rate (**Equation 2** [↗](#)) is a function of a factor M which decays in time (**Equation 3** [↗](#)), thus producing the decay in the reorientation rate. The initial reorientation rate at $t = 0$ is α .

To stochastically model both processes, we started with an initial condition of 0 reorientations ($\Omega_0 = 0$) and a large value of M to ensure a smooth decay rate ($M_0 = 1000$). Since two kinetic events are occurring (the loss of M and the gain of Ω), the algorithm simultaneously generates two time series for M and Ω , by randomly deciding when an event will occur, and the identity of that event. Two random numbers (r_1 and r_2) are drawn from a uniform distribution in $[0:1]$. One random number (r_1) is used to assign the time of an event (τ) as a combined rate of both rate processes:

$$\tau \leftarrow \frac{-\ln(r_1)}{a_0} \quad (5)$$

where:

$$a_0 = a_1 + a_2 \quad (6)$$

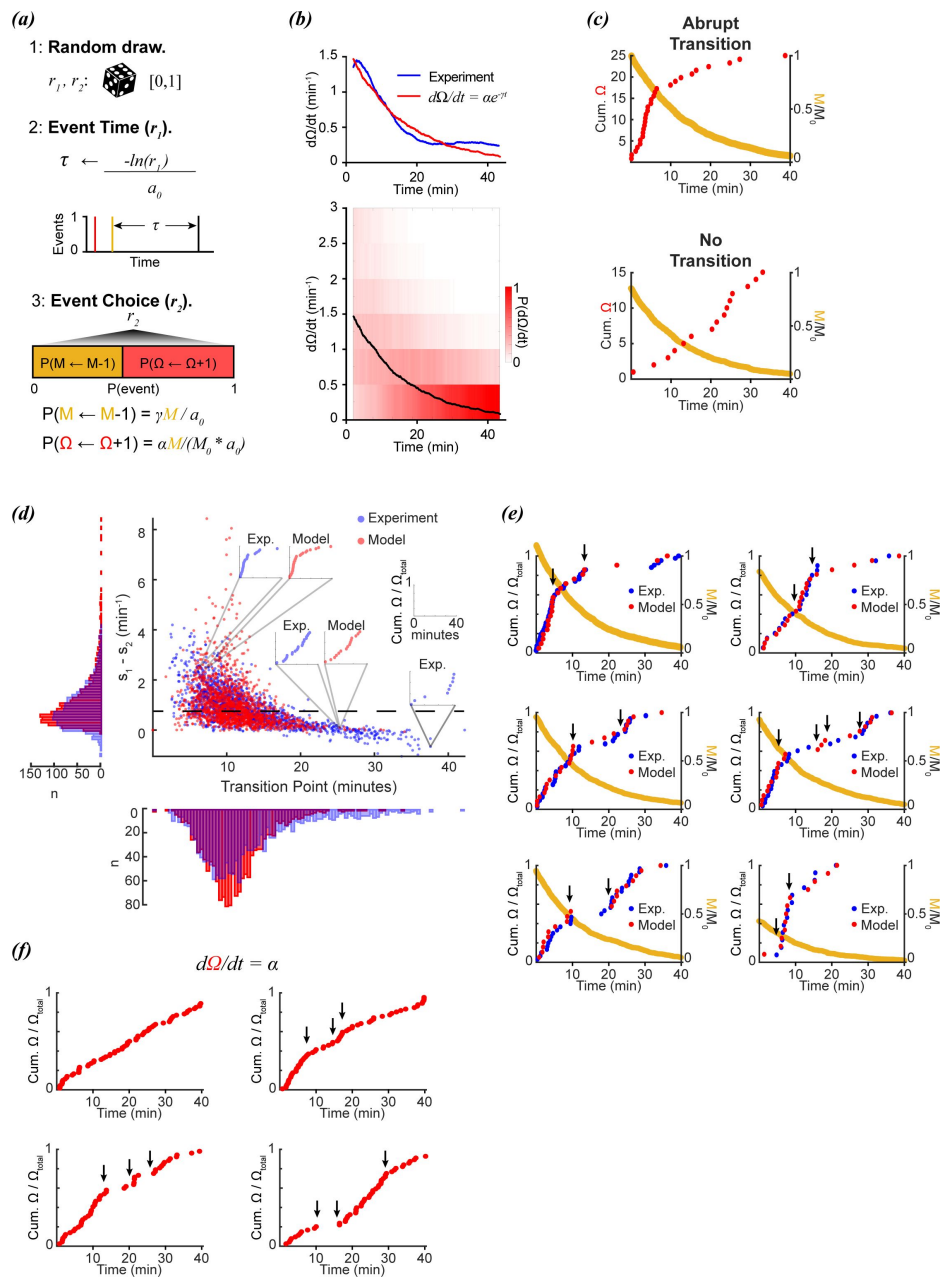


Figure 2

Stochastic modeling of foraging kinetics of *C. elegans*

(a) Outline of the Gillespie algorithm. Step 1: two random numbers (r_1 and r_2) are drawn from a uniform distribution $[0,1]$. Step 2: The time of an event is randomly assigned based on the total rates (a_0) and r_1 . Step 3: The event at $t + \tau$ is determined by the probability of the event occurring. **(b)** (Top) The parameters α and γ are assigned based on fitting a decay curve (red) to the observed average reorientation rate (blue). (Bottom) Average model population reorientation rate (black line) in a rolling 2-minute window. Red bins represent probability of observed reorientation rate. $N = 1631$, $\alpha = 1.54 \text{ min}^{-1}$, $\gamma = 0.07 \text{ min}^{-1}$. **(c)** (Top) An example of a modeled abrupt reorientation transition. (Bottom) An example of a modeled reorientation curve that lacked an abrupt reorientation transition. **(d)** Distribution of slope differences and transition times from regressions fit to the experimental (blue) and modeled (red) data. Insets are individual examples of experimental and modeled cumulative reorientation curves. Dashed line represents the median experimental slope difference. **(e)** Examples of experimental (blue) and modeled (red) cumulative reorientation curves, with similar stochastic dynamics. Sudden changes in rate are indicated with arrows. The M -decay for each model is shown in orange. **(f)** Examples of modeled data when the reorientation rate is constant. Sudden changes in rate are indicated with arrows. $\alpha = 1.5$.

A second random number (r_2) is used to calculate which event (j) ($1: \Omega = \Omega + 1$ or $2: M = M - 1$) occurs based on the relative probabilities of either event:

$$j \leftarrow \min\{n \mid r_2 \cdot a_0 \leq \sum_{i=1}^n a_i\} \quad (7)$$

For example, if the reaction rate a_1 is 10% of a_0 at time t , then there is a 10% chance that the resulting stoichiometry is due to this reaction. In this way, the Gillespie algorithm enables the simultaneous modeling of numerous reactions in parallel with machine-precision time resolution. To ensure the reorientation rate itself is a smooth decay, a large value of M is used to approximate a continuous function. The underlying assumptions are:

1. Reorientations are stochastic^{8,14}.
2. All the worms experience a common decay of a signaling factor (M) (Equation 3) that influences the reorientation rate (Equation 2).

In our approach, we fit the exponential curve in Eq. 1 to the average reorientation rate from the experimental data from López-Cruz⁵ (figure 2b), and then used these parameters (α and γ) to model 1,631 individual worms (the same number of animals in the experimental data). Starting rates were drawn from a distribution centered at α to match the observed spread of starting rates observed in the experimental data (figure 1a).

In silico reorientation curves generated with this approach produced single-transition and multi/absent transition curves, as observed experimentally (figure 2c). When plotted along with the experimental data, the *in silico* data produced a distribution of linear regression parameters comparable to the experimental worms (figure 2d). The simulation was able to produce individual trajectories that demonstrated switching behavior, despite the lack of a switching mechanism in the model (Equations 2 and 3). Furthermore, our model demonstrated a continuum of switching to non-switching behavior that was observed in experimental results (figure 2d).

The experimental transition distribution was shifted slightly earlier, however it is important to note that the experimental data were drawn from experiments where 10–15 animals were tracked together. Collisions occurred, but were excluded. However, in addition to touch, animals also reorient in response to pheromones, the presence of which may contribute to experimental observations^{21,22}. The model also deviated slightly from the experimental data for traces with late inflection points (<3% of experimental data, figure 2d). Again, since the experiments were performed with groups of animals, encounters with pheromone tracks at later time points as the animals dispersed could have potentially altered behavior^{21,22}.

This exception aside, modeling worm foraging behavior with a simple exponential decay of reorientations was sufficient to capture most experimentally observed dynamics, both switch and non-switch-like. Sudden switches between fast and slow reorientation rates were observed in both the experimental and modeled data (figure 2e), despite the model not relying on a switch strategy, and the decay in M being continuous.

An alternative search strategy is the Lévy walk, where the lengths of time between reorientations (τ) follow a bounded power-law distribution^{23,24}.

$$P(\tau) = \frac{(\mu-1)}{(a^{1-\mu}-b^{1-\mu})} \tau^{-\mu}, \quad \tau \in [a, b] \quad (8)$$

The variables a and b represent the boundaries of shortest and longest observed search lengths, respectively. In this search strategy, bouts of short path lengths randomly occur, independent of sensory information, or internal switching. While this approach also can produce sudden switches in reorientation rates, the average reorientation rate is not time-dependent, unlike what is observed in *C. elegans* (**figure 1a**). An advantage of a Lévy walk is that multiple length-scales can be sampled. Characteristically, when the distribution of path lengths is plotted on a log-log scale, the distribution is linear (**figure 3**) and “scale-free” because there is no characteristic timescale. This is unlike exponential distributions from random walks (cyan through magenta traces, **figure 3**) where the timescale is defined by the rate constant α .

There has been considerable debate about the existence of Lévy walks in foraging behavior^{25–28}, in part because observed search strategies seem to sample multiple timescales, but not to the extent of a true Lévy walk. When these foraging data are compared to only a single exponential or Lévy distribution using maximum likelihood analysis, the results tend to favor the exponential model^{27,28}, but not exclusively. Observed walk length distributions often have heavier tails than single-exponential distributions, but are not sufficiently sampled to match a Lévy walk²⁸. The distributions observed in this study from the experimental and model animals fall somewhere in between a Lévy walk and single exponential distribution (**figure 3**). Since the reorientation rate constant decays in time, the resulting distribution has a broader sampling of timescales, similar to a Lévy walk, but bounded by the initial and final observable rates. Since the reorientation rate decays in time, the breadth of this distribution increases with the total search length. This allows the animal to sample from a broader distribution of timescales than simply having one, or even two, reorientation rates.

Discussion

It is important to emphasize that the model was not explicitly designed to match the sudden changes in reorientation rates observed in the experimental data. Kinetic parameters were simply chosen to match the *average* population behavior. Sudden changes in reorientation rate were not due to sudden changes in the underlying model; stochastic behavior naturally produces sudden bunching of random events. Even if the reorientation rate is set to a constant value, stochastic sampling will still produce sudden changes in rate, even though the rate has no time-dependence (**figure 2f**).

The lack of a decision simplifies the dispersal strategy for *C. elegans*. Rather than relying on the accumulation of evidence to make a discrete decision, the worm relies on a decaying signal in the absence of food that drives the reorientation rate. This strategy increases the diffusion constant of the worm, and ensures a more efficient search strategy to find food⁴. The stochastic nature of this search is consistent with prior characterizations of worm behavior and neuronal dynamics^{8–14}, and implies that any individual worm would exhibit all the variability of the population if allowed to perform this task multiple times. We consider this to be a null model: simple stochastic models should be sufficient to explain observably stochastic behaviors.

The decay in M can be considered the memory timescale of the last food encounter. With an observed γ of 0.07 min^{-1} , this would mean a memory $\tau_{1/2}$ of ~ 10 minutes. In principle M should rise on the presence of food to increase the reorientation rate, on a timescale that is not addressed here. The reliance of the reorientation rate on a decaying factor (M) implies that the rate is influenced by a signaling factor which decays in time. It is well established that the reorientation rate is influenced by numerous signaling factors, and is the basis of the biased random walk that drives taxis in shallow gradients^{8–14}. The physical basis of M can come from multiple sources. The loss of sensory stimuli alters metabotropic glutamate signaling from sensory neurons which in turn modify the kinetics of the motor network⁵. Altered ionotropic glutamate signaling and dopamine release also influence foraging kinetics²⁹, as well as neuropeptides³⁰. The signaling

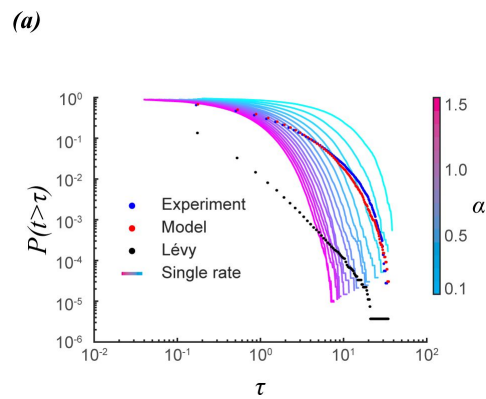


Figure 3

Scaling of foraging kinetics of *C. elegans*

(a) The cumulative distributions of run lengths for different foraging dynamics. Single rate data were generated using a single reorientation rate (α), as in [Figure 2f](#). The Lévy walk data were generated using a bounded power law distribution ([Equation 8](#), $\mu=2$), with the same boundaries as the experimental data.

factor M could be the result of one or all of these factor concentrations decaying in time. Further work will be needed to reveal how the kinetics of reorientations emerges explicitly from underlying signaling kinetics.

Data availability

Experimental data were curated from López-Cruz *et al.*⁵[↗](#).

Acknowledgements

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Additional information

Source Code availability

Analysis was performed using custom written code in MATLAB which can be found in this repository: https://github.com/GordusLab/Margolis_foraging [↗](#)

Author Contributions

A.M. and A.G. conceived the research plan, analyzed the data, and wrote the paper.

References

1. Reiss A.P., Rankin C.H. 2021) **Gaining an understanding of behavioral genetics through studies of foraging in *Drosophila* and learning in *C. elegans*** *J Neurogenet* **35**:119–131 <https://doi.org/10.1080/01677063.2021.1928113>
2. Dorfman A., Hills T.T., Scharf I. 2022) **A guide to area-restricted search: a foundational foraging behaviour** *Biological Reviews* **97**:2076–2089 <https://doi.org/10.1111/brv.12883>
3. Codling E.A., Plank M.J., Benhamou S. 2008) **Random walk models in biology** *Journal of The Royal Society Interface* **5**:813–834 <https://doi.org/10.1098/rsif.2008.0014>
4. Klein M., Krivov S.V., Ferrer A.J., Luo L., Samuel A.D., Karplus M. 2017) **Exploratory search during directed navigation in *C. elegans* and *Drosophila* larva** *Elife* **6**:e30503 <https://doi.org/10.7554/eLife.30503>
5. López-Cruz A., Sordillo A., Pokala N., Liu Q., McGrath P.T., Bargmann C.I. 2019) **Parallel Multimodal Circuits Control an Innate Foraging Behavior** *Neuron* **102**:407–419 <https://doi.org/10.1016/j.neuron.2019.01.053>
6. Calhoun A.J., Chalasani S.H., Sharpee T.O. 2014) **Maximally informative foraging by *Caenorhabditis elegans*** *eLife* **3**:e04220 <https://doi.org/10.7554/eLife.04220>
7. Killick R., Fearnhead P., Eckley I.A. 2012) **Optimal Detection of Changepoints With a Linear Computational Cost** *Journal of the American Statistical Association* **107**:1590–1598 <https://doi.org/10.1080/01621459.2012.737745>
8. Pierce-Shimomura J.T., Morse T.M., Lockery S.R. 1999) **The fundamental role of pirouettes in *Caenorhabditis elegans* chemotaxis** *J Neurosci* **19**:9557–9569 <https://doi.org/10.1523/JNEUROSCI.19-21-09557.1999>
9. Zhao B., Khare P., Feldman L., Dent J.A. 2003) **Reversal frequency in *Caenorhabditis elegans* represents an integrated response to the state of the animal and its environment** *J Neurosci* **23**:5319–5328 <https://doi.org/10.1523/JNEUROSCI.23-12-05319.2003>
10. Stephens G.J., Bueno de Mesquita M., Ryu W.S., Bialek W. 2011) **Emergence of long timescales and stereotyped behaviors in *Caenorhabditis elegans*** *Proc Natl Acad Sci U S A* **108**:7286–7289 <https://doi.org/10.1073/pnas.1007868108>
11. Flavell S.W., Pokala N., Macosko E.Z., Albrecht D.R., Larsch J., Bargmann C.I. 2013) **Serotonin and the neuropeptide PDF initiate and extend opposing behavioral states in *C. elegans*** *Cell* **154**:1023–1035 <https://doi.org/10.1016/j.cell.2013.08.001>
12. Gordus A., Pokala N., Levy S., Flavell S.W., Bargmann C.I. 2015) **Feedback from network states generates variability in a probabilistic olfactory circuit** *Cell* **161**:215–227 <https://doi.org/10.1016/j.cell.2015.02.018>
13. Roberts W.M., Augustine S.B., Lawton K.J., Lindsay T.H., Thiele T.R., Izquierdo E.J., Faumont S., Lindsay R.A., Britton M.C., Pokala N., et al. 2016) **A stochastic neuronal model predicts**

random search behaviors at multiple spatial scales in *C. elegans* *eLife* **5**:e12572 <https://doi.org/10.7554/eLife.12572>

14. Iino Yuichi, Yoshida Kazushi (2009) **Parallel Use of Two Behavioral Mechanisms for Chemotaxis in *Caenorhabditis elegans*** *J. Neurosci* **29**:5370 <https://doi.org/10.1523/JNEUROSCI.3633-08.2009>
15. Singh D., Punia B., Chaudhury S. (2022) **Theoretical Tools to Quantify Stochastic Fluctuations in Single-Molecule Catalysis by Enzymes and Nanoparticles** *ACS Omega* **7**:47587–47600 <https://doi.org/10.1021/acsomega.2c06316>
16. Srivastava N., Clark D.A., Samuel A.D.T. (2009) **Temporal Analysis of Stochastic Turning Behavior of Swimming *C. elegans*** *Journal of Neurophysiology* **102**:1172–1179 <https://doi.org/10.1152/jn.90952.2008>
17. Gillespie D.T. (1977) **Exact stochastic simulation of coupled chemical reactions** *J. Phys. Chem* **81**:2340–2361 <https://doi.org/10.1021/j100540a008>
18. Palombi F., Ferriani S., Toti S. (2020) **Coevolutionary dynamics of a variant of the cyclic Lotka–Volterra model with three-agent interactions** *The European Physical Journal B* **93**:194 <https://doi.org/10.1140/epjb/e2020-100552-5>
19. Reichenbach T., Mobilia M., Frey E. (2006) **Coexistence versus extinction in the stochastic cyclic Lotka–Volterra model** *Phys. Rev E* **74**:051907 <https://doi.org/10.1103/PhysRevE.74.051907>
20. Constable G.W.A., McKane A.J. (2015) **Models of Genetic Drift as Limiting Forms of the Lotka–Volterra Competition Model** *Phys. Rev. Lett* **114**:038101 <https://doi.org/10.1103/PhysRevLett.114.038101>
21. Hong M., Ryu L., Ow M.C., Kim J., Je A.R., Chinta S., Huh Y.H., Lee K.J., Butcher R.A., Choi H., et al. (2017) **Early Pheromone Experience Modifies a Synaptic Activity to Influence Adult Pheromone Responses of *C. elegans*** *Current Biology* **27**:3168–3177 <https://doi.org/10.1016/j.cub.2017.08.068>
22. Dal Bello M., Pérez-Escudero A., Schroeder F.C., Gore J. (2021) **Inversion of pheromone preference optimizes foraging in *C. elegans*** *eLife* **10**:e58144 <https://doi.org/10.7554/eLife.58144>
23. Humphries N.E., Sims D.W. (2014) **Optimal foraging strategies: Lévy walks balance searching and patch exploitation under a very broad range of conditions** *Journal of Theoretical Biology* **358**:179–193 <https://doi.org/10.1016/j.jtbi.2014.05.032>
24. Kölzsch A., Alzate A., Bartumeus F., de Jager M., Weerman E.J., Hengeveld G.M., Naguib M., Nolet B.A., van de Koppel J. (2015) **Experimental evidence for inherent Lévy search behaviour in foraging animals** *Proceedings of the Royal Society B: Biological Sciences* :20150424 <https://doi.org/10.1098/rspb.2015.0424>
25. Benhamou S. (2007) **HOW MANY ANIMALS REALLY DO THE LÉVY WALK?** *Ecology* **88**:1962–1969 <https://doi.org/10.1890/06-1769.1>
26. Pyke G.H. (2015) **Understanding movements of organisms: it’s time to abandon the Lévy foraging hypothesis** *Methods in Ecology and Evolution* **6**:1–16 <https://doi.org/10.1111/2041-210X.12298>

27. Edwards A.M., Phillips R.A., Watkins N.W., Freeman M.P., Murphy E.J., Afanasyev V., Buldyrev S.V., da Luz M.G.E., Raposo E.P., Stanley H.E., et al. 2007) **Revisiting Lévy flight search patterns of wandering albatrosses, bumblebees and deer** *Nature* **449**:1044–1048 <https://doi.org/10.1038/nature06199>
28. Edwards A.M., Freeman M.P., Breed G.A., Jonsen I.D. 2012) **Incorrect Likelihood Methods Were Used to Infer Scaling Laws of Marine Predator Search Behaviour** *PLOS One* **7**:e45174 <https://doi.org/10.1371/journal.pone.0045174>
29. Hills T., Brockie P.J., Maricq A.V. 2004) **Dopamine and Glutamate Control Area-Restricted Search Behavior in *Caenorhabditis elegans*** *Journal of Neuroscience* **24**:1217–1225 <https://doi.org/10.1523/JNEUROSCI.1569-03.2004>
30. Campbell J.C., Polan-Couillard L.F., Chin-Sang I.D., Bendena W.G. 2016) **NPR-9, a Galanin-Like G-Protein Coupled Receptor, and GLR-1 Regulate Interneuronal Circuitry Underlying Multisensory Integration of Environmental Cues in *Caenorhabditis elegans*** *PLOS Genetics* **12**:e1006050 <https://doi.org/10.1371/journal.pgen.1006050>

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Reviewer #1 (Public review):

Summary:

This paper concerns mechanisms of foraging behavior in *C. elegans*. Upon removal from food, *C. elegans* first executes a stereotypical local search behavior in which it explores a small area by executing many random, undirected reversals and turns called "reorientations." If the worm fails to find food, it transitions to a global search in which it explores larger areas by suppressing reorientations and executing long forward runs (Hills et al., 2004). At the population level, the reorientation rate declines gradually. Nevertheless, about 50% of individual worms appear to exhibit an abrupt transition between local and global search,

which is evident as a discrete transition from high to low reorientation rate (Lopez-Cruz et al., 2019). This observation has given rise to the hypothesis that local and global search correspond to separate internal states with the possibility of sudden transitions between them (Calhoun et al., 2014). The main conclusion of the paper is that it is not necessary to posit distinct internal states to account for discrete transitions from high to low reorientation rates. On the contrary, discrete transitions can occur simply because of the stochastic nature of the reorientation behavior itself.

Strengths:

The strength of the paper is the demonstration that a more parsimonious model explains abrupt transitions in the reorientation rate.

Weaknesses:

(1) Use of the Gillespie algorithm is not well justified. A conventional model with a fixed Δt and an exponentially decaying reorientation rate would be adequate and far easier to explain. It would also be sufficiently accurate - given the appropriate choice of Δt - to support the main claims of the paper, which are merely qualitative. In some respects, the whole point of the paper - that discrete transitions are an epiphenomenon of stochastic behavior - can be made with the authors' version of the model having a constant reorientation rate (Figure 2f).

(2) In the manuscript, the Gillespie algorithm is very poorly explained, even for readers who already understand the algorithm; for those who do not it will be essentially impossible to comprehend. To take just a few examples: in Equation (1), ω is defined as reorientations instead of cumulative reorientations; it is unclear how (4) follows from (2) and (3); notation in (5), line 133, and (7) is idiosyncratic. Figure 1a does not help, partly because the notation is unexplained. For example, what do the arrows mean, what does "*" mean?

(3) In the model, the reorientation rate $d\Omega/dt$ declines to zero but the empirical rate clearly does not. This is a major flaw. It would have been easy to fix by adding a constant to the exponentially declining rate in (1). Perhaps fixing this obvious problem would mitigate the discrepancies between the data and the model in Figure 2d.

(4) Evidence that the model fits the data (Figure 2d) is unconvincing. I would like to have seen the proportion of runs in which the model generated one as opposed to multiple or no transitions in reorientation rate; in the real data, the proportion is 50% (Lopez). It is claimed that the "model demonstrated a continuum of switching to non-switching behavior" as seen in the experimental data but no evidence is provided.

(5) The explanation for the poor fit between the model and data (lines 166-174) is unclear. Why would externally triggered collisions cause a shift in the transition distribution?

(6) The discussion of Levy walks and the accompanying figure are off-topic and should be deleted.

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Reviewer #2 (Public review):

Summary:

In this study, the authors build a statistical model that stochastically samples from a time-interval distribution of reorientation rates. The form of the distribution is extracted from a large array of behavioral data, and is then used to describe not only the dynamics of individual worms (including the inter-individual variability in behavior), but also the aggregate population behavior. The authors note that the model does not require

assumptions about behavioral state transitions, or evidence accumulation, as has been done previously, but rather that the stochastic nature of behavior is "simply the product of stochastic sampling from an exponential function".

Strengths:

This model provides a strong juxtaposition to other foraging models in the worm. Rather than evoking a behavioral transition function (that might arise from a change in internal state or the activity of a cell type in the network), or evidence accumulation (which again maps onto a cell type, or the activity of a network) - this model explains behavior via the stochastic sampling of a function of an exponential decay. The underlying model and the dynamics being simulated, as well as the process of stochastic sampling, are well described and the model fits the exponential function (Equation 1) to data on a large array of worms exhibiting diverse behaviors (1600+ worms from Lopez-Cruz et al). The work of this study is able to explain or describe the inter-individual diversity of worm behavior across a large population. The model is also able to capture two aspects of the reorientations, including the dynamics (to switch or not to switch) and the kinetics (slow vs fast reorientations). The authors also work to compare their model to a few others including the Levy walk (whose construction arises from a Markov process) to a simple exponential distribution, all of which have been used to study foraging and search behaviors.

Weaknesses:

This manuscript has two weaknesses that dampen the enthusiasm for the results. First, in all of the examples the authors cite where a Gillespie algorithm is used to sample from a distribution, be it the kinetics associated with chemical dynamics, or a Lotka-Volterra Competition Model, there are underlying processes that govern the evolution of the dynamics, and thus the sampling from distributions. In one of their references, for instance, the stochasticity arises from the birth and death rates, thereby influencing the genetic drift in the model. In these examples, the process governing the dynamics (and thus generating the distributions from which one samples) is distinct from the behavior being studied. In this manuscript, the distribution being sampled is the exponential decay function of the reorientation rate (lines 100-102). This appears to be tautological - a decay function fitted to the reorientation data is then sampled to generate the distributions of the reorientation data. That the model performs well and matches the data is commendable, but it is unclear how that could not be the case if the underlying function generating the distribution was fit to the data.

The second weakness is somewhat related to the first, in that absent an underlying mechanism or framework, one is left wondering what insight the model provides. Stochastic sampling a function generated by fitting the data to produce stochastic behavior is where one ends up in this framework, and the authors indeed point this out: "simple stochastic models should be sufficient to explain observably stochastic behaviors." (Line 233-234). But if that is the case, what do we learn about how the foraging is happening? The authors suggest that the decay parameter M can be considered a memory timescale; which offers some suggestion, but then go on to say that the "physical basis of M can come from multiple sources". Here is where one is left for want: The mechanisms suggested, including loss of sensory stimuli, alternations in motor integration, ionotropic glutamate signaling, dopamine, and neuropeptides are all suggested: these are basically all of the possible biological sources that can govern behavior, and one is left not knowing what insight the model provides. The array of biological processes listed is so variable in dynamics and meaning, that their explanation of what governs M is at best unsatisfying. Molecular dynamics models that generate distributions can point to certain properties of the model, such as the binding kinetics (on and off rates, etc.) as explanations for the mechanisms generating the distributions, and therefore point to how a change in the biology affects the stochasticity of the process. It is

unclear how this model provides such a connection, especially taken in aggregate with the previous weakness.

Providing a roadmap of how to think about the processes generating M, the meaning of those processes in search, and potential frameworks that are more constrained and with more precise biological underpinning (beyond the array of possibilities described) would go a long way to assuaging the weaknesses.

<https://doi.org/10.7554/eLife.104972.1.sa2>

Reviewer #3 (Public review):

Summary:

This intriguing paper addresses a special case of a fundamental statistical question: how to distinguish between stochastic point processes that derive from a single "state" (or single process) and more than one state/process. In the language of the paper, a "state" (perhaps more intuitively called a strategy/process) refers to a set of rules that determine the temporal statistics of the system. The rules give rise to probability distributions (here, the probability for turning events). The difficulty arises when the sampling time is finite, and hence, the empirical data is finite, and affected by the sampling of the underlying distribution(s). The specific problem being tackled is the foraging behavior of *C. elegans* nematodes, removed from food. Such foraging has been studied for decades, and described by a transition over time from 'local'/'area-restricted' search (roughly in the initial 10-30 minutes of the experiments, in which animals execute frequent turns) to 'dispersion', or 'global search' (characterized by a low frequency of turns). The authors propose an alternative to this two-state description - a potentially more parsimonious single 'state' with time-changing parameters, which they claim can account for the full-time course of these observations.

Figure 1a shows the mean rate of turning events as a function of time (averaged across the population). Here, we see a rapid transient, followed by a gradual 4-5 fold decay in the rate, and then levels off. This picture seems consistent with the two-state description. However, the authors demonstrate that individual animals exhibit different "transition" statistics (Figure 1e) and wish to explain this. They do so by fitting this mean with a single function (Equations 1-3).

Strengths:

As a qualitative exercise, the paper might have some merit. It demonstrates that apparently discrete states can sometimes be artifacts of sampling from smoothly time-changing dynamics. However, as a generic point, this is not novel, and so without the grounding in *C. elegans* data, is less interesting.

Weaknesses:

- (1) The authors claim that only about half the animals tested exhibit discontinuity in turning rates. Can they automatically separate the empirical and model population into these two subpopulations (with the same method), and compare the results?
- (2) The equations consider an exponentially decaying rate of turning events. If so, Figure 2b should be shown on a semi-logarithmic scale.
- (3) The variables in Equations 1-3 and the methods for simulating them are not well defined, making the method difficult to follow. Assuming my reading is correct, Ω should be defined as the cumulative number of turning events over time ($\Omega(t)$), not as a "turn" or "reorientation", which has no derivative. The relevant entity in Figure 1a is apparently, i.e.

the mean number of events across a population which can be modelled by an expectation value. The time derivative would then give the expected rate of turning events as a function of time.

(4) Equations 1-3 are cryptic. The authors need to spell out up front that they are using a pair of coupled stochastic processes, sampling a hidden state M (to model the dynamic turning rate) and the actual turn events, $\Omega(t)$, separately, as described in Figure 2a. In this case, the model no longer appears more parsimonious than the original 2-state model. What then is its benefit or explanatory power (especially since the process involving M is not observable experimentally)?

(5) Further, as currently stated in the paper, Equations 1-3 are only for the mean rate of events. However, the expectation value is not a complete description of a stochastic system. Instead, the authors need to formulate the equations for the probability of events, from which they can extract any moment (they write something in Figure 2a, but the notation there is unclear, and this needs to be incorporated here).

(6) Equations 1-3 have three constants (α and γ which were fit to the data, and M_0 which was presumably set to 1000). How does the choice of M_0 affect the results?

(7) M decays to near 0 over 40 minutes, abolishing ω turns by the end of the simulations. Are ω turns entirely abolished in worms after 30-40 minutes off food? How do the authors reconcile this decay with the leveling of the turning rate in Figure 1a?

(8) The fit given in Figure 2b does not look convincing. No statistical test was used to compare the two functions (empirical and fit). No error bars were given (to either). These should be added. In the discussion, the authors explain the discrepancy away as experimental limitations. This is not unreasonable, but on the flip side, makes the argument inconclusive. If the authors could model and simulate these limitations, and show that they account for the discrepancies with the data, the model would be much more compelling. To do this, I would imagine that the authors would need to take the output of their model (lists of turning times) and convert them into simulated trajectories over time. These trajectories could be used to detect boundary events (for a given size of arena), collisions between individuals, etc. in their simulations and to see their effects on the turn statistics.

(9) The other figures similarly lack any statistical tests and by eye, they do not look convincing. The exception is the 6 anecdotal examples in Figure 2e. Those anecdotal examples match remarkably closely, almost suspiciously so. I'm not sure I understood this though - the caption refers to "different" models of M decay (and at least one of the 6 examples clearly shows a much shallower exponential). If different M models are allowed for each animal, this is no longer parsimonious. Are the results in Figure 2d for a single M model? Can Figure 2e explain the data with a single (stochastic) M model?

(10) The left axes of Figure 2e should be reverted to cumulative counts (without the normalization).

(11) The authors give an alternative model of a Levy flight, but do not give the obvious alternative models:

- a) the 1-state model in which $P(t) = \alpha \exp(-\gamma t)$ (i.e. a single stochastic process, without a hidden M , collapsing equations 1-3 into a single equation).
- b) the originally proposed 2-state model (with 3 parameters, a high turn rate, a low turn rate, and the local-to-global search transition time, which can be taken from the data, or sampled from the empirical probability distributions). Why not? The former seems necessary to justify the more complicated 2-process model, and the latter seems necessary since it's the model they are trying to replace. Including these two controls would allow them to compare the number of free parameters as well as the model results. I am also surprised by the Levy model since Levy is a family of models. How were the parameters of the Levy walk chosen?

(12) One point that is entirely missing in the discussion is the individuality of worms. It is by now well known that individual animals have individual behaviors. Some are slow/fast, and similarly, their turn rates vary. This makes this problem even harder. Combined with the tiny number of events concerned (typically 20-40 per experiment), it seems daunting to determine the underlying model from behavioral statistics alone.

(13) That said, it's well-known which neurons underpin the suppression of turning events (starting already with Gray et al 2005, which, strangely, was not cited here). Some discussion of the neuronal predictions for each of the two (or more) models would be appropriate.

(14) An additional point is the reliance entirely on simulations. A rigorous formulation (of the probability distribution rather than just the mean) should be analytically tractable (at least for the first moment, and possibly higher moments). If higher moments are not obtainable analytically, then the equations should be numerically integrable. It seems strange not to do this.

In summary, while sample simulations do nicely match the examples in the data (of discontinuous vs continuous turning rates), this is not sufficient to demonstrate that the transition from ARS to dispersion in *C. elegans* is, in fact, likely to be a single 'state', or this (eq 1-3) single state. Of course, the model can be made more complicated to better match the data, but the approach of the authors, seeking an elegant and parsimonious model, is in principle valid, i.e. avoiding a many-parameter model-fitting exercise.

As a qualitative exercise, the paper might have some merit. It demonstrates that apparently discrete states can sometimes be artifacts of sampling from smoothly time-changing dynamics. However, as a generic point, this is not novel, and so without the grounding in *C. elegans* data, is less interesting.

<https://doi.org/10.7554/eLife.104972.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

This paper concerns mechanisms of foraging behavior in C. elegans. Upon removal from food, C. elegans first executes a stereotypical local search behavior in which it explores a small area by executing many random, undirected reversals and turns called "reorientations." If the worm fails to find food, it transitions to a global search in which it explores larger areas by suppressing reorientations and executing long forward runs (Hills et al., 2004). At the population level, the reorientation rate declines gradually. Nevertheless, about 50% of individual worms appear to exhibit an abrupt transition between local and global search, which is evident as a discrete transition from high to low reorientation rate (Lopez-Cruz et al., 2019). This observation has given rise to the hypothesis that local and global search correspond to separate internal states with the possibility of sudden transitions between them (Calhoun et al., 2014). The main conclusion of the paper is that it is not necessary to posit distinct internal states to account for discrete transitions from high to low reorientation rates. On the contrary, discrete transitions can occur simply because of the stochastic nature of the reorientation behavior itself.

Strengths:

The strength of the paper is the demonstration that a more parsimonious model explains abrupt transitions in the reorientation rate.

Weaknesses:

(1) Use of the Gillespie algorithm is not well justified. A conventional model with a fixed dt and an exponentially decaying reorientation rate would be adequate and far easier to explain. It would also be sufficiently accurate - given the appropriate choice of dt - to support the main claims of the paper, which are merely qualitative. In some respects, the whole point of the paper - that discrete transitions are an epiphenomenon of stochastic behavior - can be made with the authors' version of the model having a constant reorientation rate (Figure 2f).

We apologize, but we are not sure what the reviewer means by “fixed dt”. If the reviewer means taking discrete steps in time (dt), and modeling whether a reorientation occurs, we would argue that the Gillespie algorithm is a better way to do this because it provides floating-point precision time resolution, rather than a time resolution limited by dt, which we hopefully explain in the comments below.

The reviewer is correct that discrete transitions are an epiphenomenon of stochastic behavior as we show in Figure 2f. However, abrupt stochastic jumps that occur with a constant rate do not produce persistent changes in the observed rate because it is by definition, constant. The theory that there are local and global searches is based on the observation that individual worms often abruptly change their rates. But this observation is only true for a fraction of worms. We are trying to argue that the reason why this is not observed for all, or even most worms is because these are the result of stochastic sampling, not a sudden change in search strategy.

(2) In the manuscript, the Gillespie algorithm is very poorly explained, even for readers who already understand the algorithm; for those who do not it will be essentially impossible to comprehend. To take just a few examples: in Equation (1), omega is defined as reorientations instead of cumulative reorientations; it is unclear how (4) follows from (2) and (3); notation in (5), line 133, and (7) is idiosyncratic. Figure 1a does not help, partly because the notation is unexplained. For example, what do the arrows mean, what does "" mean?*

We apologize for this, you are correct, is cumulative reorientations, and we will edit the text as follows:

Experimentally, reorientation rate is measured as the number of reorientation events that occurred in an observational window. However, these are discrete stochastic events, so we should describe them in terms of propensity, i.e. the probability of observing a transitional event (in this case, a reorientation) is:

$$\frac{dP(\Omega+1,t)}{dt} = a_1 P(\Omega, t) \quad (1)$$

Here, $P(W+1,t)$ is the probability of observing a reorientation event at time t , and a_1 is the propensity for this event to occur. Observationally, the frequency of reorientations observed decays over time, so we can define the propensity as:

$$a_1 = \alpha e^{-\gamma t} \quad (2)$$

Where α is the initial propensity at $t=0$.

We can model this decay as the reorientation propensity coupled to a decaying factor (M):

$$\frac{dP(M-1,t)}{dt} = a_2 P(M, t) \quad (3)$$

Where the propensity of this event (a_2) is:

$$a_2 = -\gamma M \quad (4)$$

Since M is a first-order decay process, when integrated, the cumulative M observed is:

$$M(t) = M(0)e^{-\gamma t} \quad (5)$$

We can couple the probability of observing a reorientation to this decay by redefining (a_1 as:

$$a_1 = \frac{\alpha}{M_0} M \quad (6)$$

So that now:

$$\frac{dP(\Omega+1,t)}{dt} = \alpha e^{-\gamma t} P(\Omega, t) \quad (7)$$

A critical detail should be noted. While reorientations are modeled as discrete events, the amount of M at time $t=0$ is chosen to be large ($M_0 \leftarrow 1,000$), so that over the timescale of 40 minutes, the decay in M is practically continuous. This ensures that sudden changes in reorientations are not due to sudden changes in M , but due to the inherent stochasticity of reorientations.

To model both processes, we can create the master equation:

$$\frac{dP(\Omega, M, t)}{dt} = a_1 P(\Omega - 1, t) + a_2 P(M + 1, t) \quad (8)$$

Since these are both Poisson processes, the probability density function for a state change i occurring in time t is:

$$P(t|a_i) = a_i e^{-a_i t} \quad (9)$$

The probability that an event will *not* occur in time interval t is:

$$P(t > \tau | a_i) = \int_{\tau}^{\infty} P(t | a_i) dt = e^{-a_i \tau} \quad (10)$$

The probability that no events will occur for ALL transitions in this time interval is:

$$P(t > \tau | a_1)P(t > \tau | a_2) = \prod_i e^{-a_i \tau} = \exp[-\tau \sum_i a_i] \quad (11)$$

We can draw a random number ($r_1 \in [0,1]$) that represents the probability of no events in time interval t , so that this time interval can be assigned by rearranging equation 11:

$$\tau \leftarrow \frac{-\ln(r_1)}{a_0} \quad (12)$$

where:

$$a_0 = \sum_i a_i \quad (13)$$

This is the time interval for any event ($W+1$ or $M-1$) happening at $t + \tau$. The probability of which event occurs is proportional to its propensity:

$$P(\text{event } i) = \frac{a_i}{a_0} \quad (14)$$

We can draw a second number ($r_2 \in [0,1]$) that represents this probability so that which event occurs at time $t + \tau$ is determined by the smallest n that satisfies:

$$r_2 \cdot a_0 \leq \sum_{i=1}^n a_i \quad (15)$$

so that:

$$\text{event } i \leftarrow \min\{n \mid r_2 \cdot a_0 \leq \sum_{i=1}^n a_i\} \quad (16)$$

The elegant efficiency of the Gillespie algorithm is two-fold. First, it models all transitions simultaneously, not separately. Second, it provides floating-point time resolution. Rather than drawing a random number, and using a cumulative probability distribution of interval-times to decide whether an event occurs at discrete steps in time, the Gillespie algorithm uses this distribution to draw the interval-time itself. The time resolution of the prior approach is limited by step size, whereas the Gillespie algorithm's time resolution is limited by the floating-point precision of the random number that is drawn.

We are happy to add this text to improve clarity.

We apologize for the arrow notation confusion. Arrow notation is commonly used in pseudocode to indicate variable assignment, and so we used it to indicate variable assignment updates in the algorithm.

We added Figure 2a to help explain the Gillespie algorithm for people who are unfamiliar with it, but you are correct, some notation, like probabilities, were left unexplained. We will address this to improve clarity.

(3) In the model, the reorientation rate $d\Omega/dt$ declines to zero but the empirical rate clearly does not. This is a major flaw. It would have been easy to fix by adding a constant to the exponentially declining rate in (1). Perhaps fixing this obvious problem would mitigate the discrepancies between the data and the model in Figure 2d.

You are correct that the model deviates slightly at longer times, but this result is consistent with Klein et al. that show a continuous decline of reorientations. However, we could add a constant to the model, since an infinite run length is likely not physiological.

(4) Evidence that the model fits the data (Figure 2d) is unconvincing. I would like to have seen the proportion of runs in which the model generated one as opposed to multiple or no transitions in reorientation rate; in the real data, the proportion is 50% (Lopez). It is claimed that the "model demonstrated a continuum of switching to non-switching behavior" as seen in the experimental data but no evidence is provided.

We should clarify that the 50% proportion cited by López-Cruz was based on an arbitrary difference in slopes, and by assessing the data visually. We sought to avoid this subjective assessment by plotting the distribution of slopes and transition times produced by the method used in López-Cruz. We should also clarify by what we meant by “a continuum of switching and non-switching” behavior. Both the transition time distributions and the slope-difference distributions do not appear to be the result of two distributions. This is unlike roaming and dwelling on food, where two distinct distributions of behavioral metrics can be identified based on speed and angular speed (Flavell et al, 2009, Fig S2a). We will add a permutation test to verify the mean differences in slopes and transition times between the experiment and model are not significant.

(5) The explanation for the poor fit between the model and data (lines 166-174) is unclear. Why would externally triggered collisions cause a shift in the transition distribution?

Thank you, we should rewrite the text to clarify this better. There were no externally triggered collisions; 10 animals were used per experiment. They would occasionally collide during the experiment, but these collisions were excluded from the data that were provided. However, worms are also known to increase reorientations when they encounter a pheromone trail, and it is unknown (from this dataset) which orientations may have been a result of this phenomenon.

(6) The discussion of Levy walks and the accompanying figure are off-topic and should be deleted.

Thank you, we agree that this topic is tangential, and we will remove it.

Reviewer #2 (Public review):

Summary:

In this study, the authors build a statistical model that stochastically samples from a time-interval distribution of reorientation rates. The form of the distribution is extracted from a large array of behavioral data, and is then used to describe not only the dynamics of individual worms (including the inter-individual variability in behavior), but also the aggregate population behavior. The authors note that the model does not require assumptions about behavioral state transitions, or evidence accumulation, as has been done previously, but rather that the stochastic nature of behavior is "simply the product of stochastic sampling from an exponential function".

Strengths:

This model provides a strong juxtaposition to other foraging models in the worm. Rather than evoking a behavioral transition function (that might arise from a change in internal state or the activity of a cell type in the network), or evidence accumulation (which again maps onto a cell type, or the activity of a network) - this model explains behavior via the stochastic sampling of a function of an exponential decay. The underlying model and the dynamics being simulated, as well as the process of stochastic sampling, are well described and the model fits the exponential function (Equation 1) to data on a large array of worms exhibiting diverse behaviors (1600+ worms from Lopez-Cruz et al). The work of this study is able to explain or describe the inter-individual diversity of worm behavior across a large population. The model is also able to capture two aspects of the reorientations, including the dynamics (to switch or not to switch) and the kinetics (slow vs fast reorientations). The authors also work to compare their model to a few others including the Levy walk (whose construction arises from a Markov process) to a simple exponential distribution, all of which have been used to study foraging and search behaviors.

Weaknesses:

This manuscript has two weaknesses that dampen the enthusiasm for the results. First, in all of the examples the authors cite where a Gillespie algorithm is used to sample from a distribution, be it the kinetics associated with chemical dynamics, or a Lotka-Volterra Competition Model, there are underlying processes that govern the evolution of the dynamics, and thus the sampling from distributions. In one of their references, for instance, the stochasticity arises from the birth and death rates, thereby influencing the genetic drift in the model. In these examples, the process governing the dynamics (and thus generating the distributions from which one samples) is distinct from the behavior being studied. In this manuscript, the distribution being sampled is the exponential decay function of the reorientation rate (lines 100-102). This appears to be tautological - a decay function fitted to the reorientation data is then sampled to generate the distributions of the reorientation data. That the model performs well and matches the data is commendable, but it is unclear how that could not be the case if the underlying function generating the distribution was fit to the data.

Thank you, we apologize that this was not clearer. In the Lotka-Volterra model, the density of predators and prey are being modeled, with the underlying assumption that rates of birth and death are inherently stochastic. In our model, the number of reorientations are being modeled, with the assumption (based on the experiments), that the occurrence of reorientations is stochastic, just like the occurrence (birth) of a prey animal is stochastic. However, the decay in M is phenomenological, and we speculate about the nature of M later in the manuscript.

You are absolutely right that the decay function for M was fitted to the population average of reorientations and then sampled to generate the distributions of the reorientation data. This was intentional to show that the parameters chosen to match the population average would produce individual trajectories with comparable stochastic “switching” as the experimental data. All we’re trying to show really is that observed sudden changes in reorientation that appear persistent can be produced by a stochastic process without resorting to binary state assignments. In Calhoun, et al 2014 it is reported all animals produced switch-like behavior, but in Klein et al, 2017 it is reported that no animals showed abrupt transitions. López-Cruz et al seem to show a mix of these results, which can be easily explained by an underlying stochastic process.

The second weakness is somewhat related to the first, in that absent an underlying mechanism or framework, one is left wondering what insight the model provides. Stochastic sampling a function generated by fitting the data to produce stochastic behavior is where one ends up in this framework, and the authors indeed point this out: “simple stochastic models should be sufficient to explain observably stochastic behaviors.” (Line 233-234). But if that is the case, what do we learn about how the foraging is happening? The authors suggest that the decay parameter M can be considered a memory timescale; which offers some suggestion, but then go on to say that the “physical basis of M can come from multiple sources”. Here is where one is left for want: The mechanisms suggested, including loss of sensory stimuli, alternations in motor integration, ionotropic glutamate signaling, dopamine, and neuropeptides are all suggested: these are basically all of the possible biological sources that can govern behavior, and one is left not knowing what insight the model provides. The array of biological processes listed is so variable in dynamics and meaning, that their explanation of what governs M is at best unsatisfying. Molecular dynamics models that generate distributions can point to certain properties of the model, such as the binding kinetics (on and off rates, etc.) as explanations for the mechanisms generating the distributions, and therefore point to how a change in the biology affects the stochasticity of the process. It is unclear how this model provides such a connection, especially taken in aggregate with the previous weakness.

Providing a roadmap of how to think about the processes generating M , the meaning of those processes in search, and potential frameworks that are more constrained and with more precise biological underpinning (beyond the array of possibilities described) would go a long way to assuaging the weaknesses.

Thank you, these are all excellent points. We should clarify that in López-Cruz et al, they claim that only 50% of the animals fit a local/global search paradigm. We are simply proposing there is no need for designating local and global searches if the data don’t really support it. The underlying behavior is stochastic, so the sudden switches sometimes observed can be explained by a stochastic process where the underlying rate is slowing down, thus producing the persistently slow reorientation rate when an apparent “switch” occurs. What we hope to convey is that foraging doesn’t appear to follow a decision paradigm, but instead a gradual change in reorientations which for individual worms, can occasionally produce reorientation trajectories that appear switch-like.

As for M , you are correct, we should be more explicit. A decay in reorientation rate, rather than a sudden change, is consistent with observations made by López-Cruz et al. They found that the neurons AIA and ADE redundantly suppress reorientations, and that silencing either one was sufficient to restore the large number of reorientations during early foraging. The synaptic output of AIA and ADE was inhibited over long timescales (tens of minutes) by presynaptic glutamate binding to MGL-1, a slow G-Protein coupled receptor expressed in AIA and ADE. Their results support a model where sensory neurons suppress the synaptic output

of AIA and ADE, which in turn leads to a large number of reorientations early in foraging. As time passes, glutamatergic input from the sensory neurons decrease, which leads to disinhibition of AIA and ADE, and a subsequent suppression of reorientations.

The sensory inputs into AIA and ADE are sequestered into two separate circuits, with AIA receiving chemosensory input and ADE receiving mechanosensory input. Since the suppression of either AIA or ADE is sufficient to increase reorientations, the decay in reorientations is likely due to the synaptic output of both of these neurons decaying in time. This correlates with an observed decrease in sensory neuron activity as well, so the timescale of reorientation decay could be tied to the timescale of sensory neuron activity, which in turn is influencing the timescale of AIA/ADE reorientation suppression. This implies that our factor “M” is likely the sum of several different sensory inputs decaying in time.

The molecular basis of which sensory neuron signaling factors contribute to decreased AIA and ADE activity is made more complicated by the observation that the glutamatergic input provided by the sensory neurons was not essential, and that additional factors besides glutamate contribute to the signaling to AIA and ADE. In addition to this, it is simply not the sensory neuron activity that decays in time, but also the sensitivity of AIA and ADE to sensory neuron input that decays in time. Simply depolarizing sensory neurons after the animals had starved for 30 minutes was insufficient to rescue the reorientation rates observed earlier in the foraging assay. This observation could be due to decreased presynaptic vesicle release, and/or decreased receptor localization on the postsynaptic side.

In summary, there are two neuronal properties that appear to be decaying in time. One is sensory neuron activity, and the other is decreased potentiation of presynaptic input onto AIA and ADE. Our factor “M” is a phenomenological manifestation of these numerous decaying factors.

Reviewer #3 (Public review):

Summary:

*This intriguing paper addresses a special case of a fundamental statistical question: how to distinguish between stochastic point processes that derive from a single "state" (or single process) and more than one state/process. In the language of the paper, a "state" (perhaps more intuitively called a strategy/process) refers to a set of rules that determine the temporal statistics of the system. The rules give rise to probability distributions (here, the probability for turning events). The difficulty arises when the sampling time is finite, and hence, the empirical data is finite, and affected by the sampling of the underlying distribution(s). The specific problem being tackled is the foraging behavior of *C. elegans* nematodes, removed from food. Such foraging has been studied for decades, and described by a transition over time from 'local'/'area-restricted' search (roughly in the initial 10-30 minutes of the experiments, in which animals execute frequent turns) to 'dispersion', or 'global search' (characterized by a low frequency of turns). The authors propose an alternative to this two-state description - a potentially more parsimonious single 'state' with time-changing parameters, which they claim can account for the full-time course of these observations.*

Figure 1a shows the mean rate of turning events as a function of time (averaged across the population). Here, we see a rapid transient, followed by a gradual 4-5 fold decay in the rate, and then levels off. This picture seems consistent with the two-state description. However, the authors demonstrate that individual animals exhibit different "transition" statistics (Figure 1e) and wish to explain this. They do so by fitting this mean with a single function (Equations 1-3).

Strengths:

*As a qualitative exercise, the paper might have some merit. It demonstrates that apparently discrete states can sometimes be artifacts of sampling from smoothly time-changing dynamics. However, as a generic point, this is not novel, and so without the grounding in *C. elegans* data, is less interesting.*

Weaknesses:

(1) The authors claim that only about half the animals tested exhibit discontinuity in turning rates. Can they automatically separate the empirical and model population into these two subpopulations (with the same method), and compare the results?

Thank you, we should clarify that the observation that about half the animals exhibit discontinuity was not made by us, but by López-Cruz et al. The observed fraction of 50% was based on a visual assessment of the dual regression method we described. To make the process more objective, we decided to simply plot the distributions of the metrics they used for this assessment to see if two distinct populations could be observed. However, the distributions of slope differences and transition times do not produce two distinct populations. Our stochastic approach, which does not assume abrupt state-transitions, also produces comparable distributions. To quantify this, we will perform permutation tests on the means and variances differences between experimental and model data.

(2) The equations consider an exponentially decaying rate of turning events. If so, Figure 2b should be shown on a semi-logarithmic scale.

We are happy to add this panel as well.

(3) The variables in Equations 1-3 and the methods for simulating them are not well defined, making the method difficult to follow. Assuming my reading is correct, Ω should be defined as the cumulative number of turning events over time ($\Omega(t)$), not as a "turn" or "reorientation", which has no derivative. The relevant entity in Figure 1a is apparently $\langle \Omega(t) \rangle$, i.e. the mean number of events across a population which can be modelled by an expectation value. The time derivative would then give the expected rate of turning events as a function of time.

Thank you, you are correct. Please see response to Reviewer #1.

(4) Equations 1-3 are cryptic. The authors need to spell out up front that they are using a pair of coupled stochastic processes, sampling a hidden state M (to model the dynamic turning rate) and the actual turn events, $\Omega(t)$, separately, as described in Figure 2a. In this case, the model no longer appears more parsimonious than the original 2-state model. What then is its benefit or explanatory power (especially since the process involving M is not observable experimentally)?

Thank you, yes we see how as written this was confusing. In our response to Reviewer #1, we added an important detail:

While reorientations are modeled as discrete events, which is observationally true, the amount of M at time $t=0$ is chosen to be large ($M_0 \leftarrow 1,000$), so that over the timescale of 40 minutes, the decay in M is practically continuous. This ensures that sudden changes in reorientations are not due to sudden changes in M , but due to the inherent stochasticity of reorientations.

However you are correct that if M was chosen to have a binary value of 0 or 1, then this would indeed be the two state model. Adding this as an additional model would be a good idea to compare how this matches the experimental data, and we are happy to add it.

(5) Further, as currently stated in the paper, Equations 1-3 are only for the mean rate of events. However, the expectation value is not a complete description of a stochastic system. Instead, the authors need to formulate the equations for the probability of events, from which they can extract any moment (they write something in Figure 2a, but the notation there is unclear, and this needs to be incorporated here).

Thank you, yes please see our response to Reviewer #1.

(6) Equations 1-3 have three constants (alpha and gamma which were fit to the data, and M0 which was presumably set to 1000). How does the choice of M0 affect the results?

Thank you, this is a good question. We will test this down to a binary state of M as mentioned in comment #4.

(7) M decays to near 0 over 40 minutes, abolishing omega turns by the end of the simulations. Are omega turns entirely abolished in worms after 30-40 minutes off food? How do the authors reconcile this decay with the leveling of the turning rate in Figure 1a?

Yes, reviewer #1 recommended adding a baseline reorientation rate which is likely more biologically plausible. However, we should also note that in Klein et al they observed a continuous decay over 50 minutes.

(8) The fit given in Figure 2b does not look convincing. No statistical test was used to compare the two functions (empirical and fit). No error bars were given (to either). These should be added. In the discussion, the authors explain the discrepancy away as experimental limitations. This is not unreasonable, but on the flip side, makes the argument inconclusive. If the authors could model and simulate these limitations, and show that they account for the discrepancies with the data, the model would be much more compelling. To do this, I would imagine that the authors would need to take the output of their model (lists of turning times) and convert them into simulated trajectories over time. These trajectories could be used to detect boundary events (for a given size of arena), collisions between individuals, etc. in their simulations and to see their effects on the turn statistics.

Thank you, we will add error bars and perform a permutation test on the mean and variance differences between experiment and model over the 40 minute window.

(9) The other figures similarly lack any statistical tests and by eye, they do not look convincing. The exception is the 6 anecdotal examples in Figure 2e. Those anecdotal examples match remarkably closely, almost suspiciously so. I'm not sure I understood this though - the caption refers to "different" models of M decay (and at least one of the 6 examples clearly shows a much shallower exponential). If different M models are allowed for each animal, this is no longer parsimonious. Are the results in Figure 2d for a single M model? Can Figure 2e explain the data with a single (stochastic) M model?

Thank you, yes, we will perform permutation tests on the mean and variance differences in the observed distributions in figure 2d. We certainly don't want the panels in Figure 2e to be suspicious! These comparisons were drawn from calculating the correlations between all model traces and all experimental traces, and then choosing the top hits. Every time we run the simulation, we arrive at a different set of examples. Since it was recommended we add a baseline rate, these examples will be a completely different set when we run the simulation, again.

We apologize for the confusion regarding M . Since the worms do not all start out with identical reorientation rates, we drew the initial M value from a distribution centered on M_0 and a variance to match the initial distribution of observed experimental rates.

(10) The left axes of Figure 2e should be reverted to cumulative counts (without the normalization).

Thank you, we will add this. We want to clarify that we normalized it because we chose these examples based on correlation to show that the same types of sudden changes in search strategy can occur with a model that doesn't rely on sudden rate changes.

(11) The authors give an alternative model of a Levy flight, but do not give the obvious alternative models:

a) the 1-state model in which $P(t) = \alpha \exp(-\gamma t) dt$ (i.e. a single stochastic process, without a hidden M , collapsing equations 1-3 into a single equation).

b) the originally proposed 2-state model (with 3 parameters, a high turn rate, a low turn rate, and the local-to-global search transition time, which can be taken from the data, or sampled from the empirical probability distributions). Why not? The former seems necessary to justify the more complicated 2-process model, and the latter seems necessary since it's the model they are trying to replace. Including these two controls would allow them to compare the number of free parameters as well as the model results. I am also surprised by the Levy model since Levy is a family of models. How were the parameters of the Levy walk chosen?

Thank you, we will remove this section completely, as it is tangential to the main point of the paper.

(12) One point that is entirely missing in the discussion is the individuality of worms. It is by now well known that individual animals have individual behaviors. Some are slow/fast, and similarly, their turn rates vary. This makes this problem even harder. Combined with the tiny number of events concerned (typically 20-40 per experiment), it seems daunting to determine the underlying model from behavioral statistics alone.

Thank you, yes we should have been more explicit in the reasoning behind drawing the initial M from a distribution (response to comment #9). We assume that not every worm starts out with the same reorientation rate, but that some start out fast (high M) and some start out slow (low M). However, we do assume M decays with the same kinetics, which seems sufficient to produce the observed phenomena.

(13) That said, it's well-known which neurons underpin the suppression of turning events (starting already with Gray et al 2005, which, strangely, was not cited here). Some discussion of the neuronal predictions for each of the two (or more) models would be appropriate.

Thank you, yes we will add Gray et al, but also the more detailed response to Reviewer #2.

(14) An additional point is the reliance entirely on simulations. A rigorous formulation (of the probability distribution rather than just the mean) should be analytically tractable (at least for the first moment, and possibly higher moments). If higher moments are not obtainable analytically, then the equations should be numerically integrable. It seems strange not to do this.

Thank you for suggesting this, we will add these analyses.

*In summary, while sample simulations do nicely match the examples in the data (of discontinuous vs continuous turning rates), this is not sufficient to demonstrate that the transition from ARS to dispersion in *C. elegans* is, in fact, likely to be a single 'state', or this (eq 1-3) single state. Of course, the model can be made more complicated to better match the data, but the approach of the authors, seeking an elegant and parsimonious model, is in principle valid, i.e. avoiding a many-parameter model-fitting exercise.*

*As a qualitative exercise, the paper might have some merit. It demonstrates that apparently discrete states can sometimes be artifacts of sampling from smoothly time-changing dynamics. However, as a generic point, this is not novel, and so without the grounding in *C. elegans* data, is less interesting.*

Thank you, we agree that this is a generic phenomenon, which is partly why we did this. The data from López-Cruz seem to agree in part with Calhoun et al, that claim abrupt transitions occur, and Klein et al, which claim they do not occur. Since the underlying phenomenon is stochastic, we propose the mixed observations of sudden and gradual changes in search strategy are simply the result of a stochastic process, which can produce both phenomena for individual observations.

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